
At the mercy of the mountain? Field stations and the culture of volcanology

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Abstract. This paper discusses the role of volcano observatories as local sites of knowledge production through the monitoring of volcanoes, and links this with the global production of knowledge in volcanology. The paper initially discusses the institutional geography of volcanological science, and relates this broadly to the geography of science and science studies. It then focuses on interview data to draw some insights into the experiences of observatory and university scientists. Observatories have to provide scientific advice to local authorities and also engage with the international scientific community. They may act as gatekeepers to the field, and as rich sources of baseline data for research projects. As such, they have a pivotal role in the production of knowledge beyond the field sites—and are also involved in the protection of populations. The paper argues that the tension between academic research and volcano monitoring that occurs at many observatories is an important part of the process of knowledge production in volcanology, and also reveals the complex geographies of social responsibility and collaboration that exist around active volcanoes as scientific networks and territorial governance structures interact.

Keywords: field stations, volcanology, volcanic risk, science and policy, geographies of science

Introduction

Volcanoes have been described as ‘laboratories’ (eg, Bonaccorso et al, 2004), places where scientists can make significant breakthroughs—particularly, perhaps, during eruptions (eg, Donovan et al, 2012a). They may also be of cultural and historical significance—for example, the ancient and modern histories of Vesuvius (eg, Cocco, 2013)—and sources of legend (eg, Cashman and Cronin, 2008; Oppenheimer, 2011). Many social scientific studies of volcanic risk have tended to focus on the populations around volcanoes, rather than on the scientific community that studies them—yet volcano observatories are in many places key stakeholders and participants in the complex geographical narratives that surround volcanoes. Observatories are responsible to local and national governments for provision of advice about volcanic activity, and so must engage with the population and local authorities—but they also engage with the wider scientific community. They have civil responsibilities and act as channels for scientific data collection. In so doing, they engage both with territorial governance structures and with global knowledge networks.

In this context, the 2012 conviction for multiple manslaughter passed down to six scientists and a local official in an Italian court, for failing to communicate the risk prior to the devastating L’Aquila earthquake, is of critical concern to volcanologists (Marzocchi, 2012). The volcanological community was urged by leading scientists to check on local laws concerning litigation and scientific advice (IAVCEI, 2012). There was concern about the potential for volcano observatories and scientific advisors to find themselves in similar situations—though it is worth noting that the events in L’Aquila were perhaps particularly complex, and that the accusation related not to the integrity of the science, but to the communication of the risk (see, for example, Alexander, 2013). Both volcanologists and

seismologists deal with very high levels of uncertainty in their work, and the implications for society may be severe. Volcanologists can detect signals from magma at depth, but interpreting the data and translating it to a forecast is challenging (Donovan et al, 2012b; Tilling, 2008). In addition, low-probability, high-impact risks are difficult to manage both scientifically and politically (Government Office for Science, 2011). The precautionary principle is not always an obvious response to low-probability events, since the impact of evacuations can be economically severe and lead to undesirable consequences including outbreaks of infectious disease and social unrest. This can lead to very high pressures on scientists working at the policy interface. The response of the International Association of Volcanology and Chemistry of the Earth's Interior (IAVCEI; the professional organisation for volcanologists) to the L'Aquila verdict was perhaps more condemnatory even than many responses from seismologists (IAVCEI, 2012), revealing the magnitude of anxieties felt by practitioners. The nature of knowledge and uncertainty, and how these are used and communicated in volcanic risk management, cannot be readily separated from ideas of social responsibility and accountability in democracies. The case of L'Aquila shows the contestation of the role of scientists providing advice to governments under high uncertainty. It demonstrates the importance of understanding the social context of volcanology and its ongoing discourses. IAVCEI's reaction also shows the complexity of interactions between scientific networks and national governments and legal systems.

In this paper we discuss the geographical characteristics of knowledge production at volcano observatories, using data from interviews with scientists and local officials. Initially, we frame our discussion in the broader disciplinary context of 'geography of science' and science and technology studies (STS). We then discuss the global networks and institutional variation between countries, before focusing on interview data from Montserrat to look at particular issues in more detail. Drawing on ideas from the geographies of environmental governance (eg, Bulkeley, 2005; Massey, 2004), the paper employs two concepts to explore the negotiation of local and global scales through the juxtaposition of different kinds of network: the geography of responsibility (which links to government and policy on multiple scales, some dictated by the nation-state) and the geography of collaboration (interaction within the scientific community around the world). It will show that the geographically specific nature of knowledge acquisition at volcano observatories derives from the social context of the science both within and beyond the practice of science itself, but that this knowledge also feeds into much larger and more centralised debates about the science of volcanoes and the social and political roles of scientists. Furthermore, the potentially awkward juxtaposition of topologies—political responsibilities and scientific networks—both enhances and complicates the work that observatories do.

Volcanoes around the world: geography of science and its practices

This section introduces some key issues about knowledge production in volcanology in relation to place and culture. It initially discusses the role of place in the context of scientific knowledge more broadly, and then provides some examples of the dialogue between local and global in volcanological knowledge production and its social context. It makes some observations about institutional geography: the networks and relationships that define observatories, both in terms of their responsibility and their relationship with volcanological knowledge networks.

The role of place and social context in the production of scientific knowledge has been discussed widely in both geography and STS (eg, Livingstone, 2003; Massey, 1992, 1999; Powell, 2007; Shapin, 1995; Whatmore, 2009). Science is embodied in its practitioners and audiences, and is located not only in its institutions but also in sites of discovery in the field (eg, de Bont, 2009; Dewsbury and Naylor, 2002; Harrison et al, 2004; Kohler, 2002; Livingstone, 2003;

Naylor, 2002; Powell, 2002; 2008). This opens up the possibility of a 'geography of science', which brings together geography and STS (Doubleday 2007; Livingstone, 2003; Massey, 1999; Shapin, 1998; Whatmore, 2009), and links both to physical aspects of the science and to sociological accounts of scientific lives (eg, Collins, 1985; Traweek, 1988). Indeed, geography defines many scientific narratives—stories of scientific exploration, for example, by their nature unite the geographical with the scientific enterprise (Withers et al, 2006; Larson, 2011). Field stations, too, act as gateways to nature and interfaces between science and policy (eg, Wråkberg, 2013). Astronomical observatories, for example, function as thresholds through which scientists have access to the stars (eg, Hoskin, 1999), while meteorological stations have responsibilities for public information and documentation at multiple scales (eg, Naylor, 2006). At volcano observatories, scientists and populations negotiate a complex, impressive, and yet potentially dangerous geography. These interactions depend on the physical state of the volcanic system, the institutional and political frameworks in place for monitoring volcanoes, and the discourses within scientific research more broadly. They take place around dynamic, changing landscapes that represent a complicated dialogue between humanity, human networks, and nature (Massey, 2006). This feeds into the culture of volcanology.

The idea of a 'culture' is contested in the social sciences; in this paper it denotes the shaping of groups of individuals by practices, knowledges, discourses, and languages. Scientific disciplines can exhibit cultural attributes through their use of language and collaborative working in the acquisition of knowledge through human actors and networks. There are also local cultures—within institutions—and larger ones, as academics, for example, take part in broader discourses. There is a geographical exchange between intensely local knowledges that are locally applied in risk assessment on individual volcanoes—which could be described as geographical 'anchors' (Haggett and Chorley, 1969)—and the application of these ideas to more general models of volcanic activity via continuously changing networks of institutions and personnel. Ultimately, the scientific practices that result from this interaction are also dependent on the nature, duration, and intensity of the volcanic activity itself: volcanic eruptions increase the rate of knowledge production in volcanology, and are critical in its advancement (eg, Donovan et al, 2012a). Anecdotes from archetypal eruptions have become abundant in volcanological pedagogy, and represent an important interface between locally embedded knowledge and the global discourse of volcanic risk management. There is a complex intertwining of the geography of volcanic risk and that of volcanological science: stories about risky situations in particular places feed into the development of the discipline.

One example of this uncertain negotiation of scale concerns the comparability of volcanoes: to what extent can the behaviour of one volcano be indicative of behaviour at another? This is a geographical problem of induction: apparently similar activity at two volcanoes may have very different consequences—thus making the act of extrapolation from one case study to the next problematic. For example, in 1976 La Soufrière de Guadeloupe reawakened, prompting a controversial evacuation of 72 000 people for several months, at high economic cost (Fiske, 1984). The crisis did not escalate to the point of a major eruption. In 1995 Soufrière Hills Volcano on Montserrat displayed very similar initial activity, and then erupted episodically for 15 years. This posed a challenge early in the crisis, though scientists discerned the escalating activity in time to evacuate local people (Aspinall et al, 2002). Individual volcanoes can experience a range of magnitudes of eruption and produce a range of products. The assumption that future eruptions will be similar to past eruptions is also questionable. Making judgments about scale, timing, and nature of eruptions can be highly uncertain, and, while there are generalisations about global activity that can be made, these may provide only a very qualified contribution to decision making during specific volcanic crises—and assuming that populations will respond in

the same way is also flawed (eg, Chester and Duncan, 2007). However, there is abundant evidence that preparation is critical prior to volcanic eruptions (eg, Chester et al, 2002), as societies may become heavily dependent on scientific advice overnight (eg, Donovan et al, 2013). Thus, while volcanoes are geographical anchors, they are changeable and dynamic, and can force considerable topological stretching and reshaping upon human social and political networks (see Amin, 2011; Jones, 2009; Massey, 1999; 2006)—including those of scientific advice and knowledge production.

The link between scientific knowledge production and societal benefit has been raised in the STS literature (Beck, 1992; Gibbons et al, 1994; Jasanoff, 2004; 2005; Nowotny et al, 2001). One common formulation refers to blue-skies research as ‘mode 1’ and socially applied science as ‘mode 2’. Mode-2 science suggests that scientists have a duty to serve society, but is less clear about how or by whom that service should be defined, or about the role of institutional and geographical context on defining priorities for research. The points raised about volcanology above show that knowledge production in volcanology may depend on local human and physical geographies and responsibilities, but also reaches beyond them into the broader scientific community through connections such as narratives and fieldwork. This paper aims to draw out some of these themes empirically. While much of the literature on the geography of science is based within historical accounts (Livingstone 1995; 2003; 2005; Naylor, 2002; Naylor et al, 2008; Powell, 2007), this paper is concerned with the very recent past—1995 to 2009—and so its methods include qualitative interviews as well as documentary analysis. It also analyses ongoing discourses, and demonstrates the relevance of geographical studies of science for understanding the geography of volcanic risk (Donovan and Oppenheimer, 2012). The next section of the paper will describe some of the international differences between observatories, and discuss their broad role, before moving on to present empirical data from field studies.

The geography of observatories around the world: research and monitoring

The sites of volcanological practice are diverse: the places that volcanology inhabits are complex not only in their human geography, but in their physical geography. Volcanology is practised in the field, at volcano observatories and other monitoring organisations, and in a purely academic, often university-based environment. There is considerable overlap of personnel—many volcanologists will work in both contexts during their careers—but the fact that careers may be judged on the basis of publications can create frustration for long-term observatory scientists who spend long periods of time processing data, and therefore have less time for writing papers. This is exacerbated when papers are published by researchers concerning volcanic activity that the observatory had recorded but not had time to publish on [Donovan and Oppenheimer (2012); see also Newhall et al (1999) who present a protocol for volcanologists during eruptions]. At the same time, observatories may be poorly funded, and have difficulty justifying the purchase of new equipment; this can create opportunities and tensions in collaborative ventures: researchers developing new technologies may wish to test them at active volcanoes, and in doing so wish to collaborate with observatories. Observatories thus serve as research gateways. This makes them comparable with polar field stations (eg, Bravo, 2006; Powell, 2007; 2008), but also extends into the area of social responsibility in crisis situations with very high stakes. In response to this, observatories have to develop scientific practices that protect them from liability and ensure the safety of citizens, as well as engaging actively with the research community. Observatory culture must look inwards to the scientific community, and outwards to the social and political context, via a range of social interactions. For example, relationships between technical staff who are locally based and scientists from overseas can be critical in facilitating scientific research.

Observatories may also contribute significantly to local realisations of governmentality: since evacuation decisions and land-use planning are heavily dependent on scientific advice, observatories may be ‘blamed’ for unpopular political decisions that seek to manage a population (Donovan et al, 2013). While the official line may be that the scientists are available purely for advice, in practice they may be taking decisions as well—particularly if an alert-level system is directly tied to particular social actions such as evacuations—and if the decisions are based solely on their advice, the public perception may be that the scientists are responsible (Haynes, 2005; Pattullo, 2000). A further concern for scientists at observatories is how far they are responsible for educating the local population and engaging in outreach activities. Thus, political context and institutional structure are important.

There are a range of factors that affect the location and management of volcano observatories, and considerable variation between nations in how such institutions are organised and incorporated into decision making. For example, the Institut de Physique du Globe du Paris manages the French volcanoes centrally, but has local field stations with a few scientists close to the volcanoes (eg, Guadeloupe, Martinique, Reunion). In the event of volcanic unrest, further personnel can be dispatched. In Italy, the Osservatorio Etneo (OE) in Catania is a large institution that has responsibility for Etna, Stromboli, and Vulcano (with a small field station on Stromboli). OE is part of the Istituto Nazionale per la Geofisica e Vulcanologia, which has divisions in many Italian cities and also runs the Osservatorio Vesuviano. Most observatories are run by geological surveys or similar institutions, but there is variation in the location of the observatory itself. Some countries monitor volcanoes remotely, often as part of the remit of a national geological survey, using both ground-based and satellite methods. Some of these countries also have small stations on the volcanoes themselves, particularly if certain volcanoes are very active. Volcano monitoring at observatories is tied in via networks to the Volcanic Ash Advisory Centres, which have responsibility for monitoring ash hazard for aviation purposes and are typically based in meteorological stations. A detailed discussion of this is beyond the scope of this paper, but following the 2010 Eyjafjallajökull eruption, additional pressure was placed on governments to monitor their volcanoes locally (in 2004, Ewert and Newhall found that less than 5% of active volcanoes were adequately monitored). However, this depends considerably on resources, particularly in countries like Chile and Indonesia, which have numerous volcanoes, and in poor countries, where the resources to monitor volcanoes at all are lacking. This is the case for many African countries (Lenhardt and Oppenheimer, 2014).

The configuration of volcano observatories thus depends on several factors: the organisation of scientific work and civil protection within a country; the presence of a geological survey; the number of volcanoes (physical geography) and the risk associated with particular volcanoes—which may in turn depend upon observed historical activity and recent monitoring; and financial constraints. These issues underlie the practical running of volcano observatories. They also constitute complex topologies that interact with one another, as civil and government bodies are linked with local institutions that collaborate internationally with research-dominated institutions such as universities. In the next sections, data from interviews with scientists and local officials are presented. These data were collected on Montserrat, but involved interviews with scientists who had experience of working in a range of contexts. They allow some insights into the geographies of responsibility and collaboration within and beyond volcanic settings.

Methods

The fieldwork for this paper was undertaken in 2008–09 and involved participant observation (Geertz, 1973) and semistructured interviews with scientists and local officials on Montserrat, a UK Overseas Territory in the Eastern Caribbean, where volcanic eruptions occurred episodically from 1995 to 2010 (Druitt and Kokelaar, 2002; Fergus, 2004). Montserrat Volcano

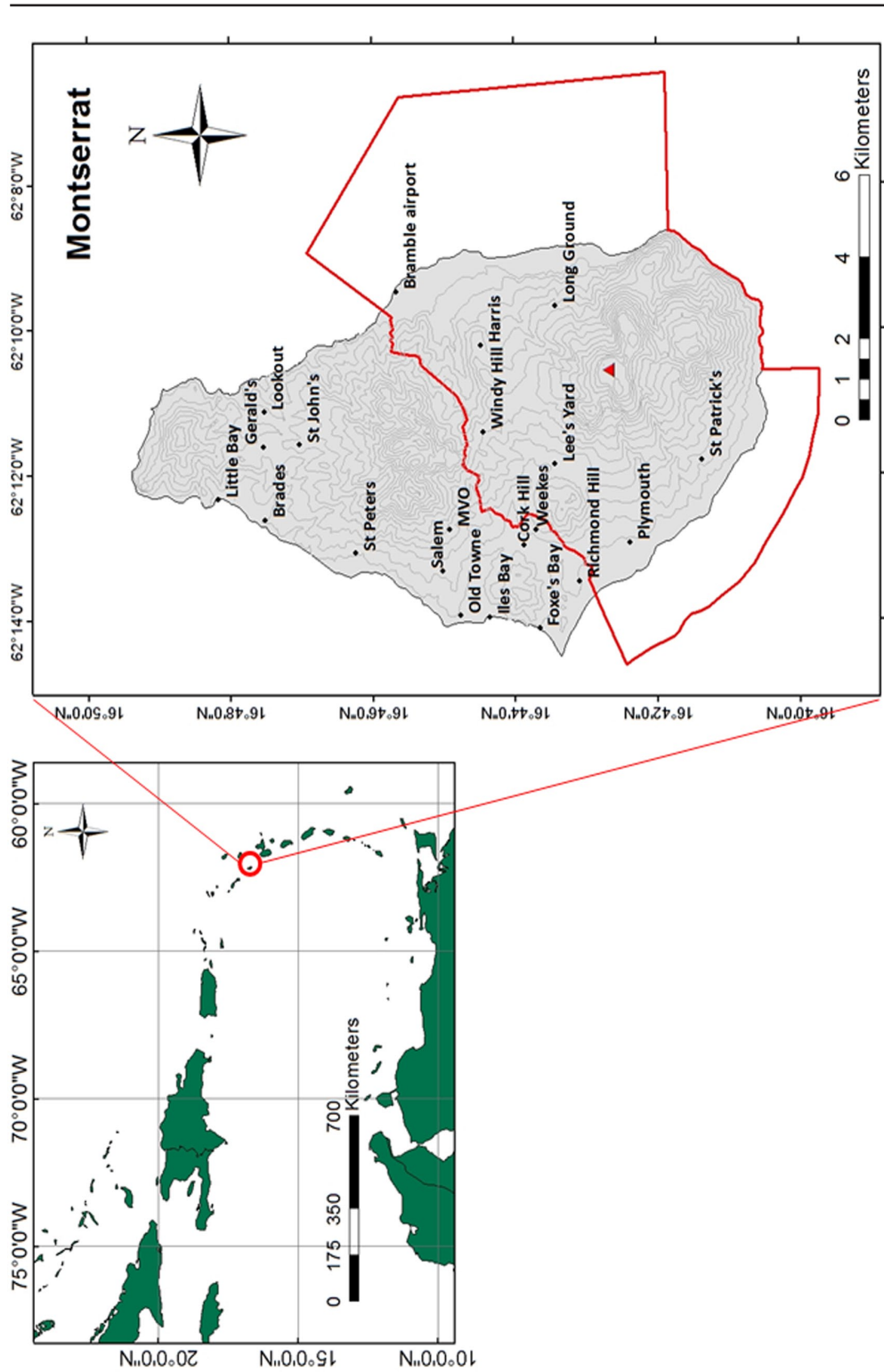


Figure 1. [In colour online.] Map showing the location of Montserrat in the Lesser Antilles Volcanic Arc, and a map of the island showing the permanent exclusion zone in red. The volcano is represented by a red triangle. Contours are taken from an SRTM Digital Elevation Model, and are at 50 m intervals. Note the location of the Montserrat Volcano Observatory, within line of sight of the volcano.

Observatory (MVO) was established to monitor the volcano and advise the authorities on its activity. It works in conjunction with a Scientific Advisory Committee (SAC), which was meeting every six months during the period covered in this paper. The volcanic eruptions on Montserrat forced the abandonment of the capital city, Plymouth, in 1996, and, combined with associated lahars, have subsequently destroyed it (figure 1).

Interviews ranged in length from 30 minutes to 2 hours. In total, sixty-two interviews were carried out, forty-one with elites and twenty-one with local residents. Interviewees were selected on the basis of their occupation (elites) or by snowball sampling (residents). Interview data were coded thematically, and the data presented below are representative of the entire dataset. An indication of the role of each scientist at the time of interview is given as this is relevant to the argument, but other information is withheld to preserve anonymity. There were two field seasons, each ten weeks in duration. It should be noted that scientists interviewed for this project, though at the time of interview working for or with the MVO, have between them experience of observatories in France, New Zealand, Papua New Guinea, the Philippines, Indonesia, the USA, and Vanuatu. Additional participant observation was carried out at the Osservatorio Etneo in 2008, and these data feed into the argument below.

A variety of narratives and opinions was expressed during the fieldwork, both from interviews and from participant observation. The perspectives of university and observatory scientists, clerical staff, and local officials and publics were considered. It should be noted that different institutions have different mandates, resources, and responsibilities, so the information given in this section should not be taken as universally representative or as the view of a single institution. However, it is likely to have echoes beyond the fieldwork locations. Volcano observatories manage a middle ground between locally specific and sited knowledge, and the much broader global volcanological knowledge base.

Results and discussion

Policy, research, and social responsibility

Volcano observatories (taken here to include all institutions with responsibility for volcano monitoring under a civil authority) are the sites of baseline knowledge acquisition and dissemination, initially within the scientific community, but also far beyond it. They have to balance their civil protection responsibilities with the pressures for academic research and career development, and must work with other researchers elsewhere in the development and testing of new technologies, both for monitoring purposes and for the furtherance of volcanological knowledge. These conflicts of interest and responsibility raise a number of general issues about the processes involved in knowledge acquisition: scientists at volcano observatories have to balance their bifurcated role [monitoring and research; see also Burt (1994) on the interdependence of research and monitoring in environmental management], with prioritising research and developing and testing new technologies. In addition, their role may be defined by both the type of volcano they work on, and by the interests of local, regional, and wider communities.

When interviewed, scientists raised several aspects of observatory management, ranging from their daily activities to their role in governance. The driving force behind the work of observatories was universally recognised:

“You have lives to protect ... you’re trying to help the community as well, so you don’t want anyone to be down Salem and sleeping if there’s something happening, like over the weekend or during the evenings” (technician, observatory).

This suggests that social responsibility changes the nature of the scientific endeavour: there is a drive to collect long time-series of consistent data in order to answer social questions about

safety, and also a need to maintain equipment and respond to emergency alarms. This differs from research in universities, which generally lacks the immediacy:

“The biggest crime is if you stop collecting data and this is different for universities—they collect data for specific things; we basically want continuous data because that tells you when things are changing, and that requires a different way of working Leading edge technology doesn’t work in an observatory. Trailing edge technology does—it’s tried and proven and doesn’t cause you any problems” (senior scientist, observatory).

The primary aim of providing protection is not, however, the only aim of observatories. Observatory data can be used retrospectively to make significant scientific advances (eg, Voight et al, 1999). Additionally, observatories frequently have a role in facilitating university research. The idea of observatories as data acquisition centres, and the relationship of this role to research, was discussed at length, and many quotations in this section refer to it. The need to keep the monitoring network as a separate installation was repeatedly stressed—rather than integrating it with researchers’ networks, which might be differently configured but also used by the observatory. One of the challenges for observatory managers in the first instance, then, is to convince the funders—often local governments—that research is a valid function for an observatory.

In its early years the MVO had to educate its governmental funders so that they understood the relationship between monitoring and research in volcanology: it is not a true dichotomy (Clay et al, 1999; Donovan et al, 2013). At the same time, there were people on Montserrat whose houses had been destroyed, so funding scientific luxuries was clearly out of the question. Attaining a balance between research and monitoring was a moral issue as well as an operational one. One of the questions that arose from this was how observatories integrate research into the monitoring process:

“I think the people who are collecting the data and doing the monitoring and doing the hard work . . . should really have an opportunity to work on the research associated with it The key thing is the monitoring, but it’s good for the observatory to do research too” (senior scientist, university; observatory experience).

Research in an observatory requires time and motivation. It is regarded as important but also has to be weighed up in terms of its financial cost. This may be particularly challenging in a small observatory with limited resources. This issue extends the realm of the observatory director: as well as working with interactional expertise between disciplines (Collins and Evans, 2007), observatory directors have to be aware of legal, economic, and political contexts—requiring a transdisciplinary expertise. There is also the need to interact with the public:

“You should have an outreach officer who should ideally have a scientific background but they work alongside you, maybe even having a low-level role in the team as a scientist, which could step up at times of crisis” (senior scientist, observatory experience).

Furthermore, the nature of science is such that there are often multiple interpretations of the same dataset, and this requires judgment when information is being used to advise policy makers.

“The job . . . does stop you doing science . . . the job is providing advice and information and that takes up the whole of your time I think it’s vital that a scientist is doing that because . . . there are always areas where [there are] questions and differences in interpretation” (senior scientist, observatory).

The social role of the observatory is thus twofold: providing bottom-up education for the public, and providing high-level advice to governments and local officials. The observatory is socially embedded, both within the local community, and within the wider scientific community as it collaborates with external scientists, and it can be affected by the perceptions of both groups concerning its work.

Interviewees related the impulse to do research to the environment within the observatory as well as beyond it. A problem on Montserrat is its size—"the other issue really is the isolation, so you need observatories that are big enough to have a big team" (scientist, observatory)—not possible on a small island. Several interviewees mentioned scientific discussions as a key part in promoting scientific advancement—the 'internal dynamics of exchange'—which is dependent on personalities as well as management styles. One senior scientist (observatory) summed up thus:

"[X] was saying, 'well, you can see this relationship and this correlation in the papers I'm writing ...' and I was like, 'Show it now, so we can discuss it, and if you have some hot stuff that you want to keep privy, then don't mention it, but divulge a little bit, and don't just say, well my job is to do the counting and to do the processing', because then it's too much of a monitoring attitude In an observatory, you have to do monitoring ... that answers specific questions; you have to acquire data that will allow questions to be answered in ten years, five years, by other people—data mining. You also have to do monitoring in the sense that you test new methods that people from the outside come and test ... they don't take too much of your time, they do it with their own resources, they give you full access to the data and full understanding of the limitations of the data, ... and you have to ... say, OK that's very interesting but I cannot commit time and integrate this yet ... but do it, keep us informed ... so you have to manage these three types of monitoring—short term, long term, and then prototype, avant-garde monitoring."

This is a long quotation, but it demonstrates several important points directly. It also shows the value of perspective: looking at the work of the observatory over different periods of time *and also in the global context of volcanic risk*, with a broader responsibility to inform risk assessment elsewhere. There are several projects, also mentioned by the interviewee, that are seeking to maximise the global relevance of data from individual volcanoes (eg, WOVODat, Earth Observatory of Singapore), and contextualising a single field site within this network allows for an additional sense of purpose, both for monitoring and for undertaking some in-house research: "We need to think about what data is necessary and what are the questions that we need to answer so that data-mining can go on in an observatory" (scientist, observatory). These comments imply that the issues are fundamentally a matter of individual philosophies: why people do volcanology. Similarly, monitoring and research seem to lie on a spectrum rather than express a dichotomy, and often come down to an individual's philosophy and attitude—their willingness to work overtime in a crisis, for example (Donovan et al, 2012a). The commitment to public safety thus runs through observatory work, and entails communication skills:

"You have to do a fantastic job with all the science and all of that but you have to be able to communicate the science not only to the people but also to the authorities, and communicate it in such a way that they have confidence in what you're saying" (senior scientist, university; observatory experience).

While publication of scientific papers is seen as important, the maintenance of authority, continuity, and public trust may be a more immediate stake at observatories. The challenges arise when the two come into competition, suggesting that the idea of monitoring and research as a spectrum might be a helpful conceptual model.

It would be tempting to apply the ideas of mode-1 and mode-2 science from STS (Gibbons et al, 1994; Nowotny, 2003; Nowotny et al, 2001) to research and monitoring, respectively. However, as demonstrated above, this is neither accurate nor helpful. In this context the epistemological position of the two modes of working is best characterised geographically. Monitoring, in its *primary purpose*, is locally embedded. Research can transcend the local—both in its subject matter and in the networks that result from it via journals and academic discourse. Monitoring is driven by the geography of risk, and the *responsibility* to find and

manage the human limits of that risk. Research can feed into monitoring, but is fundamentally far broader in both its purposes and its application. In the next section, we examine some of the challenges that arise from the interaction of these geographies in institutional contexts.

Frontier field stations: collaboration between universities and observatories

Collaboration may take several forms—such as externally driven research projects, or outsourced monitoring projects and observatory-led research, an example of which would be petrological data analysis carried out for an observatory that lacks the necessary equipment. The distinction between outsourcing/offshoring and externally driven research can be important, not least in terms of financial management. If the research is driven from the outside but useful to the observatory, it can be very complicated. This is particularly the case where new potential monitoring technologies are being developed:

“If it’s someone just experimenting I don’t want to spend thirty or forty thousand dollars for them to figure out if it really works, it might be much better to collaborate with somebody else who has a similar instrument and see what they think will work for us down the road before actually making the purchase” (local official, finance).

While of interest to the observatory, such “avant-garde” monitoring is particularly valued by researchers who wish to test their ideas and technologies—and the sharing of academic resources with observatories was valued highly. However, there is also a caution attached to this: observatory scientists and visiting researchers have very different day-to-day priorities. This can create tension between observatory scientists and those who visit briefly to gather data, relying on the observatory for logistical support.

Observatory scientists mentioned that university scientists were sometimes unaware or unappreciative of the amount of work done by observatory scientists, not least to protect them when they go into unsafe areas:

“If you’re going to go into an unsafe area we’re going to have to have staff in the operations room to make radio contact with you, so no matter how you look at it, you’re going to involve the observatory, and you have to have some sort of arrangement” (local official, local government).

Asked about collaboration, administrators felt that once the work is done it is beneficial, but the financial side can be complicated. An example on Montserrat is when collaborators send instruments to the MVO, the MVO pays the customs charge and collects the item—and then has to reclaim the money from the collaborators, which takes time and resources. Also time consuming is the process of monitoring itself:

“There’s a lot of tensions created in a volcanic crisis related to data and the differences between academic research and monitoring work—working in an observatory ... it’s a generic problem. Obviously people working in observatories gathering data and monitoring the volcano often don’t have time to really analyse the data very carefully. They sometimes feel that they’re left out” (senior scientist, university; observatory experience).

This was reflected in comments from observatory scientists that academic researchers did not always view them as equals, and also that observatories can feel isolated geographically from the scientific community. However, the idea that there are valuable alternative perspectives from doing work in observatories was discussed by a number of interviewees in relation to the act of interpreting data when a social role is being undertaken. This comes back to the fact that volcanology is still a young science with high levels of uncertainty, and measuring that uncertainty is problematic, especially with regard to the nuances involved in the interpretative process. This suggests a different type of volcanology prevails in observatories:

“At a university you’re very concentrated on your own bit of data, and to you it’s the most interesting thing in the world, and it tells you everything and you don’t realise (a) there’s

other data that might conflict with it and (b) your interpretation might be wrong” (senior scientist, observatory).

This subjective view of the scientific process suggests that the epistemic practices of observatories are thought to provide a broader analysis of the volcano than the more specific research campaigns. The focus on publication and career clearly also creates problems when university researchers and observatory scientists collaborate:

“They are trying to get published, they’re trying to get their name out there, they’re trying to get a job that can pay them, and so in some ways I guess you have to try and understand that, but when they come, if they come with the wrong attitude, then you’re less willing to help” (technician, observatory).

There are several facets, then, to the question of career development at observatories. While the ‘pressure to publish’ acts as a ‘control mechanism’ for integrity, it can also lead to the isolation of those who have to spend a lot of time monitoring and do not have time for research. This in turn can prove problematic when it comes to the release of monitoring data:

“They had all this data that they wanted to write up ... but they didn’t want to release it to anybody else, so you ended up with this enormous gap of data where it wasn’t written up, nobody had seen it, nobody knew what it was, nobody knew what it showed, and ... it’s quite dangerous” (senior scientist, observatory).

Several interviewees discussed or hinted at moral implications for the way that scientists operate when carrying out research on active volcanoes:

“There’s cases ... where an individual scientist is very protective about their data or their ideas, and not terribly open-minded, and therefore that creates difficulties because ... you can’t wait for somebody to publish their paper before looking at the data, so in academic circles it might be reasonable to say ... I don’t want to disclose this data until I’ve published my paper, I think that’s fine. If they sit on it for 20 years, that’s not fine and again I think that’s part of the moral protocol in science” (senior scientist, university; observatory experience).

This was a topic covered by the IAVCEI Crisis Protocols (Newhall et al, 1999), which was also discussed with the interviewees.

“I read it a long time ago and I think I was happy with it ... I treat it more as a document for people from outside observatories that come to work at observatories, to say ‘please don’t mess with what we’re doing’. ‘Cos that’s where I saw problems” (senior scientist, observatory).

Other scientists noted that the inclusion of observatory scientists on papers, even as tenth author or lower, did at least show their involvement in the science, some of which would not be possible without their monitoring efforts. In terms of career development, having a relatively meagre list of publications is not necessarily a disadvantage for applying for other observatory or monitoring jobs: “in some ways, it’s changing the thoughts of the employers” (senior scientist, observatory), the awareness of people at a distance from observatory work. However, this may also depend on the local context—some scientists work in observatories for their entire career:

“The challenge is to be able to maintain discussions with colleagues that are purely in the academic world, from far away—that’s not easy, because often people come for their fieldwork, they do the work and then they leave and then there’s the paper ... and it’s difficult to maintain the exchange ... So I think scientific research, a little bit of research, has to be done in house or you have to have a setup such that observatory scientists can take time off every year ... to go to a conference, go to a specific workshop” (senior scientist, observatory).

The problem with this setup is that “when you leave the island, you implode because you’ve been at such high activity that you just want to sit back and do nothing. You need to be very motivated” (senior scientist, observatory).

Over the course of their career, many scientists move between observatories and universities: there is no clear-cut divide, and there is considerable knowledge exchange, largely as a result of initiatives from within the scientific community—yet opportunities for observatory scientists to take a sabbatical may be very limited. It may also be very difficult for scientists to reenter the university system, if there are stringent publication requirements (for example, in the UK). However, the various commissions of the IAVCEI have deliberately sought to foster good feelings and expertise transfer between different communities:

“Scientists who collaborate across the ‘divide’ develop personal friendships that promote ongoing dialogue and generation of ideas for proposals, etc. The plight of well-trained volcanologists in underresourced contexts is quite compelling for a comfortable university scientist with any degree of humanity!” (senior scientist, university).

The importance of friendship was mentioned by several interviewees—personal relationships and successful collaborations in the past clearly play a very significant role in directing future plans, not only for projects but also for potential career moves. Many projects are based on networks. The act of embedding oneself in the broader culture of volcanology is mutually beneficial. There are a range of contexts for this—field visits by university scientists, international conferences, and e-mail contact. E-mail between the MVO and university academics on the Scientific Advisory Committee and beyond was noted by interviewees as a critical tool in the management of the eruption.

There was also a concern that visiting scientists not push aside the organisations given the responsibility to monitor the volcano. In spite of the challenges, however, there was a consensus among scientists that collaborations should be encouraged:

“I think that cooperation with universities or other research institutions might be helpful. I visited an observatory ... and they are extremely badly funded, but nevertheless they do very good work because they get all the equipment through cooperations and they have simple memoranda of understanding and this is how they operate” (senior scientist, university).

Observatory scientists may be working more closely with scientists from other disciplines for risk assessments, which may provide them with a level of interactional expertise that surpasses that of academics who do not make an effort to move outside their field. Observatories by their purpose require that their scientists operate beyond the boundaries of individual disciplines, as well as seeking to contribute to them. Multidisciplinarity is inherent in observatory work—and is the focus of many of the current research projects mentioned above. The transfer of ideas between observatories and universities—and the transfer of personnel—shows that not only are research and monitoring not dichotomous, they are also mutually sustaining. Observatories are thus laboratories both for the generation and translation of scientific data and research, and for the seeding of public policy.

Conclusions

It follows from the data presented above that there is a complex relationship between space, epistemology, and risk in volcanology. The distinction made by Gibbons et al (1994) between mode-1 and mode-2 science breaks down in this case, because monitoring and research feed into each other and cannot be defined purely by location: observatories do most of the monitoring, but they also do research. In addition, more and more research is mode-2 driven, since this has become important to funding bodies and to some scientists. While the distinction between the theory of science and its practice is more effective, since most theoretical work is done in universities, it fails to account adequately for the social

responsibility that observatory work entails—or for its complex relationship with space. The human geography of volcanology incorporates both scientists and populations via the geography of volcanic risk, which has its origins in the physical but which is characterised by complex connectivities both within science (between observatories and university scientists, for example) and beyond it (as observatories relate to social and political structures).

Livingstone (2003) identified three aspects of the relationship between science and space: the site, region, and circulation of scientific people, knowledge, and objects. He argued that the geography of science is crucial in understanding the history and nature of scientific knowledge. Recent debates in human geography concerning the nature of space–time and relationality can take this further (eg, Bulkeley 2005; Jones, 2009; Massey, 2004; Paasi, 2011): space is never neutral, but evolves on multiple scales and in many directions. The notion of topology is useful here, because it allows for the ways in which networks can be deformed and altered through time and space (eg, Paasi, 2011). In this paper we argue that the geography of science in volcanology is dictated by the human and physical geographies of volcanic risk: the geography of responsibility that governs the observatory and its managing organisation (where one exists), and the geography of collaboration, which allows the sharing of resources and knowledges between field stations, central institutions, and universities. The latter moves in two directions: theoretical geophysical studies can benefit from empirical data from field stations, and field stations may benefit from new models and methods. They may also benefit through the secondment of trained personnel in a crisis, or the temporary involvement of external institutions (such as the United States Geological Survey Volcanic Disasters Assistance Programme, which provides advice ‘behind the scenes’ to volcanologists in other nations).

Thus, while the volcano itself is a physical point that can be conceived of as a ‘geographical anchor’, the topologies that relate to it adjust both to its dynamism in space and time, and to the changing demands of social and scientific institutions: volcanoes are also assemblages of human and physical entities, ideas, and networks (Donovan and Oppenheimer, 2015). Furthermore, the broader scientific community, with its interests in making comparison between volcanoes and identifying and defining archetypes, extends beyond the ‘local’. The multiple temporal and spatial scales on which volcanology has to operate feed into its nature and into the relationships between scientists and the subjects/objects that they study. Volcanoes are constant reminders of human limitations, as well as provocative of study and knowledge production (eg, Massey, 2006). Governments bear responsibility for the management of volcanic risk—and may exercise governmentality through the application of scientific advice, involving scientists in the outworking of power dynamics at a territorial level. The management of observatories may thus be characterised by their explicit role in the protection of citizens and the practices of local and national governments. The experience of interviewees from Montserrat [admittedly complicated by the island’s colonial government structures (Donovan et al, 2013)] and elsewhere was that the structures and practices of government had a major impact on the nature of scientific working, from the provision of equipment to the interpretation of results. While research grants may increasingly have to demonstrate social impacts, the dependence of observatories on *government practice* (in relation both to government structure and to the beliefs held by government officials regarding science, risk, and the role of scientists) was referred to repeatedly. Observatories simultaneously have to deal with the state (often an ostensibly vertical relationship), citizens (through, for example, friendships and outreach), and with the scientific networks of volcanology. Interaction with governments and citizens is dominated by ideas of responsibility, while that around volcanology is associated with collaborative knowledge production.

Knowledge in volcanology may bear ‘the imprint of its location’ (Livingstone, 2003), but it also frequently transcends location, becoming part of the geographical problem of induction that pervades volcanic risk assessment: the nature and limits of similarity between

places are uncertain. The geography of volcanic risk thus feeds into the geography of science, complicating it and even on occasion providing a significant moral context. It is relational: it is dependent on the building of relationships between collaborators and technicians, observatory scientists and university scientists and also on the broader topologies of power and knowledge. Observatories are hubs for informal networks of scientific practitioners working with the aim of understanding the local natural system better and producing cutting-edge research that will feed into future interpretations of the risk from volcanic activity. Individual volcanoes therefore affect—and are affected by—developments in knowledge production and in governance. They are material manifestations of human dialogues that transcend scale through networks whilst also focusing on it in intensely practical ways.

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